

Gradient Descent and Learning-Rate Control

A model improves by taking useful downhill steps.

NAME _____

ROLL NO. _____

DATE _____

What you will be able to do

1. perform gradient descent from scratch on a simple loss function;
2. compare Vanilla (full-batch) GD, SGD, and Mini-batch GD;
3. diagnose a fixed learning rate from its behaviour; and
4. choose and explain a learning-rate schedule.

How to use this worksheet

Read the short explanation, inspect the visual or worked example, and then attempt each boxed task before checking with your instructor. The goal is not to memorise names: it is to predict what an optimiser will do and explain why.

Part 1 Finding downhill

1.1 The idea behind gradient descent

Imagine walking down a foggy hill while blindfolded. You cannot see the lowest point, but you can feel the local slope under your feet. A sensible strategy is to take a small step in the direction that slopes most sharply downward. Training a model uses the same idea.

Key idea - the update rule

Let $L(\theta)$ be the loss and θ be the model parameter(s). The gradient $\nabla L(\theta_t)$ points uphill, so we move in the opposite direction:

$$\theta_{t+1} = \theta_t - \alpha \nabla L(\theta_t)$$

Here $\alpha > 0$ is the **learning rate**: it controls the size of each step. The gradient controls the direction.

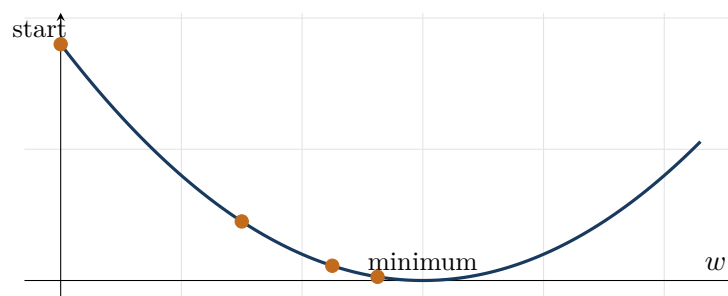


Figure 1 - A one-parameter loss. The orange points are successive gradient-descent updates moving toward the bottom.

1.2 Gradient descent from scratch

Worked example Let $L(w) = (w - 3)^2$, so $\frac{dL}{dw} = 2(w - 3)$. Start at $w_0 = 0$ and use $\alpha = 0.25$.

$$\begin{aligned} g_0 &= 2(0 - 3) = -6, & w_1 &= 0 - 0.25(-6) = 1.5, & L(w_1) &= (1.5 - 3)^2 = 2.25 \\ g_1 &= 2(1.5 - 3) = -3, & w_2 &= 1.5 - 0.25(-3) = 2.25, & L(w_2) &= 0.5625 \end{aligned}$$

The gradient is negative while $w < 3$, so subtracting it increases w . As we approach the minimum, the slope and the update both shrink.

Your turn Complete two more updates

Using $w_2 = 2.25$ and $\alpha = 0.25$:

$$g_2 = 2(\text{_____} - 3) = \text{_____} \quad w_3 = \text{_____} - 0.25(\text{_____}) = \text{_____}$$

Is $L(w_3)$ smaller than $L(w_2)$? Explain in one sentence.

Your turn Turn the mathematics into a training loop

Complete the three missing expressions. Read each line aloud: *measure loss, measure slope, step downhill*.

```
w = 0.0
alpha = 0.25
for step in range(4):
    loss = _____
    grad = _____
    w = w - _____
    print(step, w, loss)
```

If you changed α from 0.25 to 1.2, what would you expect to happen near $w = 3$?

Part 2 Three ways to ask the data

2.1 Vanilla Gradient Descent - ask the whole class

Suppose every training example is a student giving an opinion about which way is downhill. **Vanilla Gradient Descent**, also called **full-batch GD**, asks *every* student before taking one step. Its gradient is accurate and smooth, but expensive when the class has millions of students.

Vanilla Gradient Descent

For N examples, compute one exact average gradient before each update:

$$\nabla L(\theta) = \frac{1}{N} \sum_{i=1}^N \nabla L_i(\theta), \quad \theta \leftarrow \theta - \alpha \nabla L(\theta).$$

It makes **one update per epoch**. An epoch is one complete pass through the training data.

Your turn *Fill in the consequence*

Full-batch GD has a _____ (noisy / accurate) gradient. For $N = 50,000$ examples, each update requires seeing _____ examples. This makes each update _____ (cheap / expensive).

2.2 Stochastic Gradient Descent - ask one student

Stochastic Gradient Descent (SGD) asks only one randomly selected student, then updates immediately. The answer is noisy: one example may not represent the whole dataset. However, it is cheap and provides many frequent updates.

SGD

At step t , choose one example i and use its gradient:

$$\theta \leftarrow \theta - \alpha \nabla L_i(\theta).$$

With N training examples, true SGD makes N updates per epoch. The noisy steps can help it explore rather than follow one perfectly smooth path.

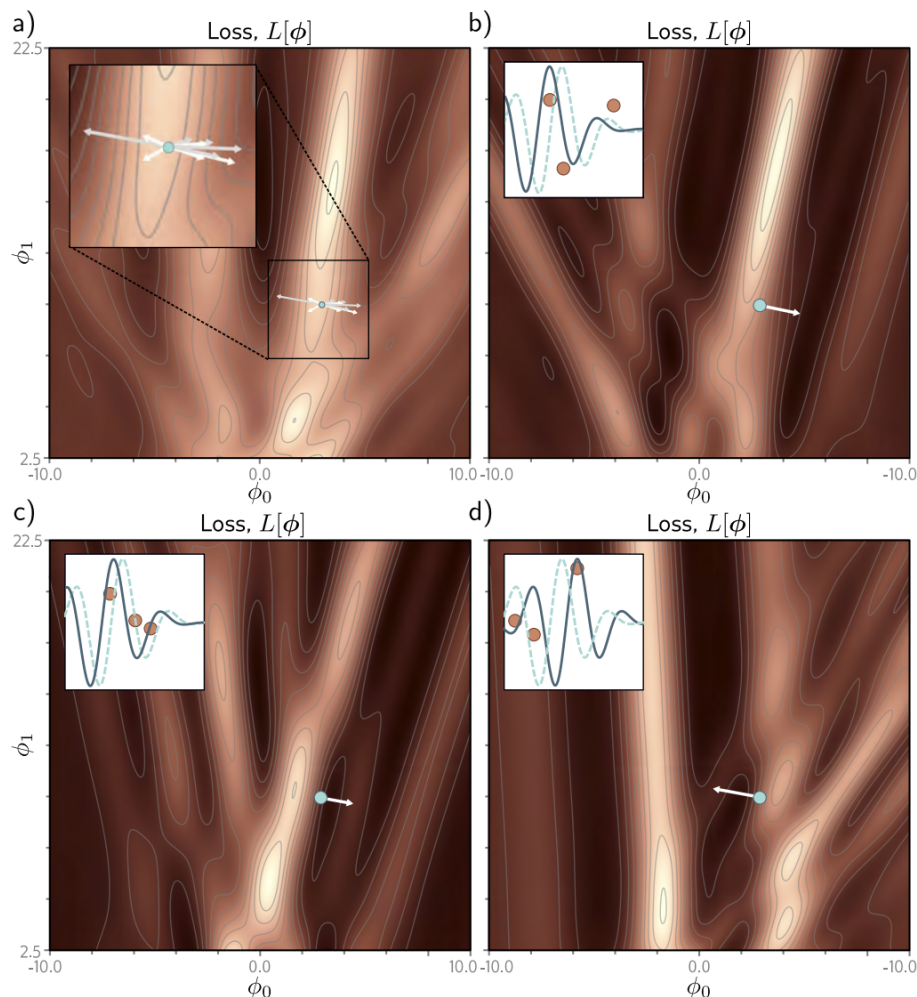


Figure 2 - A noisy stochastic trajectory. Different sampled examples produce slightly different local slopes, so the path wiggles while moving downhill on average. Source: existing course visual adapted from Prince, *Understanding Deep Learning*, Ch. 6.

2.3 Mini-batch Gradient Descent - ask a small group

Mini-batch Gradient Descent asks a small random group, such as 32 or 64 examples. It keeps much of SGD's speed while averaging away some of its noise. In modern deep learning, people often casually say "SGD" even when they mean mini-batch SGD.

Mini-batch Gradient Descent

For a batch of B examples:

$$\widehat{\nabla L}(\theta) = \frac{1}{B} \sum_{i \in \mathcal{B}} \nabla L_i(\theta), \quad \theta \leftarrow \theta - \alpha \widehat{\nabla L}(\theta).$$

With N examples, one epoch contains N/B updates.

Your turn *Compare the three estimators*

Method	Examples per update	Gradient noise	Updates per epoch
Vanilla / full-batch GD	all N	_____	_____
True SGD	_____	high	_____
Mini-batch GD	B	_____	N/B

You have a GPU and 1,000,000 images. Which method is the practical default, and why?

Part 3 Choosing the size of every step

3.1 Fixed Learning Rate

A **fixed learning rate** uses the same α at every update. It is simple, but one step size must serve two different phases: travelling quickly when far away, and settling precisely when close to the minimum.

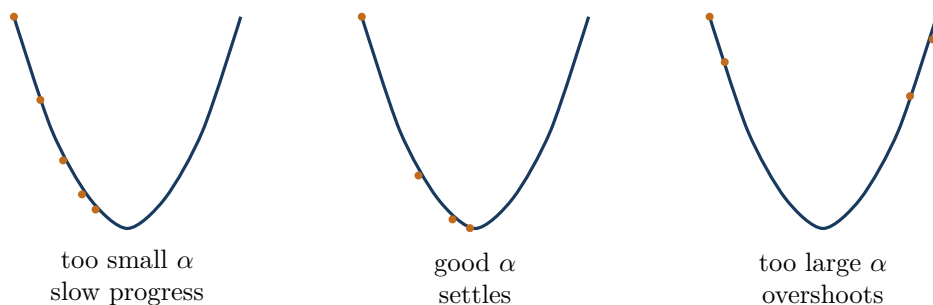


Figure 3 - The same bowl, three step sizes. A suitable rate makes visible progress without repeatedly crossing the minimum.

Your turn *Diagnose the behaviour*

Write one item from the word bank beside each observation.

Word bank: *stagnation* *oscillation* *divergence*

Observation	Diagnosis
Loss falls very slowly for many epochs.	_____
Loss goes down overall, but bounces up and down near the minimum.	_____
Loss rises rapidly; values may become NaN.	_____
For each diagnosis, is α too low, slightly too high, or much too high?	

A practical way to choose a starting rate

An **learning-rate range test** starts with a tiny rate and gradually increases it while observing the loss. Choose a value before the loss begins to rise sharply. This gives a defensible starting point; validation loss then tells you whether it generalises.

Part 4 Accelerate, then brake**4.1 Learning Rate Scheduling**

Driving at full speed is useful on an open highway but not while parking. A **learning-rate schedule** makes α large enough for early progress and smaller later for careful settling. Instead of a constant, α becomes α_t .

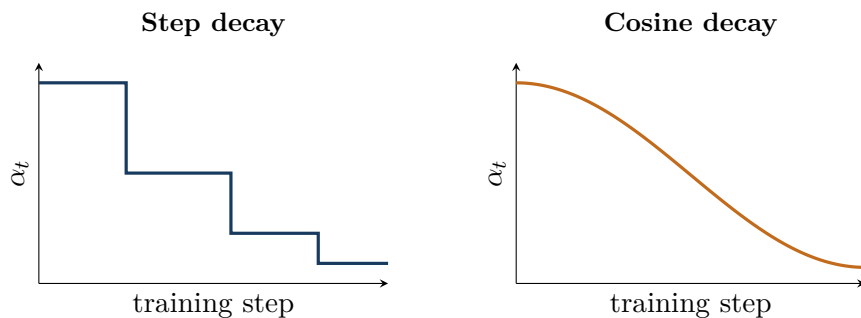


Figure 4 - Two time-driven schedules. Step decay drops suddenly; cosine decay makes a smooth landing.

Common schedules

- **Step decay:** keep α constant, then reduce it by a fixed factor at milestones.
- **Exponential decay:** reduce smoothly by the same percentage over time.
- **Cosine decay:** gently lower α following the upper half of a cosine curve.
- **Warm-up:** start small and increase α for the first few steps to avoid unstable early updates.
- **Reduce on plateau:** watch validation loss and reduce α only when progress stalls.

Your turn *Match each schedule to its behaviour*

Write A–E in the left column.

Schedule		Behaviour
Step decay	___	A. Starts near zero and rises for a short initial period.
Exponential decay	___	B. Reduces only after validation performance stops improving.
Cosine decay	___	C. A smooth percentage-style fall.
Warm-up	___	D. A smooth soft landing shaped like half a cosine.
Reduce on plateau	___	E. A staircase of sudden reductions.

Your validation loss has not improved for five epochs, but you do not know how many total epochs you will train. Which schedule is a natural fit? Why?

Practice and assessment**5.1 Match the following**

Draw lines or write the correct letter. Each description is used once.

Term		Description
1. Vanilla Gradient Descent	___	A. Uses a small subset to estimate the gradient.
2. Stochastic Gradient Descent	___	B. One step size is used throughout training.
3. Mini-batch Gradient Descent	___	C. Uses all examples before updating.
4. Fixed Learning Rate	___	D. Updates after one randomly selected example.
5. Reduce on Plateau	___	E. Lowers the rate when validation progress stalls.

5.2 MCQs

Circle one answer for each question. Do not use notes for the first attempt.

MCQ 1. If the gradient is positive, $\theta \leftarrow \theta - \alpha \nabla L$ moves θ in which direction?

- A. positive B. negative C. random D. zero

MCQ 2. Which method uses all N training examples for one gradient estimate?

- A. Vanilla GD B. true SGD C. mini-batch GD D. warm-up

MCQ 3. With $N = 1,000$ and $B = 50$, a mini-batch epoch contains:

- A. 1 update B. 20 updates C. 50 updates D. 1,000 updates

MCQ 4. Which gradient estimate is usually noisiest?

- A. full-batch B. mini-batch of 128 C. true SGD D. cosine decay

MCQ 5. A loss that explodes upward most strongly suggests:

- A. α is too low B. α is too high C. batch size is zero D. gradient is exact

MCQ 6. What is the main purpose of mini-batches?

- A. remove all noise B. balance cost and gradient noise C. use no data D. avoid updates

MCQ 7. What does warm-up do?

- A. begins with a smaller rate and increases it B. always halves the rate
C. uses the whole dataset D. stops training early

MCQ 8. Which schedule reacts to validation loss rather than a predefined epoch?

- A. step decay B. exponential decay C. reduce on plateau D. cosine decay

MCQ 9. Why might a rate that was useful early become harmful late?

- A. gradients become smaller near the minimum B. data disappears
C. epochs are forbidden D. the loss becomes constant everywhere

MCQ 10. Cosine decay is best described as:

- A. sudden staircase drops B. a smooth soft landing C. one random rate D. an increase forever

5.3 Subjective Questions

Q1. For $L(w) = (w - 4)^2$, start at $w_0 = 0$ with $\alpha = 0.25$. Compute g_0 and w_1 . Show the substitution.

Q2. You are training on 2 million images with a GPU. Choose full-batch GD, true SGD, or mini-batch GD, and justify your choice using both cost and noise.

Q3. Explain how noisy SGD updates can sometimes be useful early in training, but troublesome near a minimum.

Q4. Training and validation loss both fall very slowly. Diagnose the learning-rate issue and state one intervention.

Q5. A model has unstable early updates and a known training budget. Propose a schedule and explain each phase.
